$\mu = 62.75/75$, $\sigma = 6.48$, $\text{Med} = 63$

ECE 329 | Fields and Waves I | Summer 18

Kudeki

Exam 3

Thu, July 25, 2018 — 11 AM - 11:50

Name:	Solution
Section:	

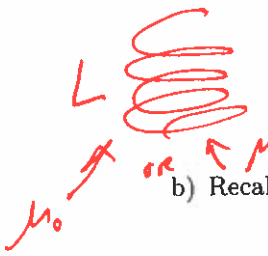
Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the above boxes.

This is a closed book exam and calculators are allowed. You are allowed three formula sheets of 8.5 by 11 inch dimensions — both sides of the sheets may be used. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate.

Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	

1. This problem has two independent parts:

   a) (10 pts) I have a rod of some solid with an unknown permeability μ. To determine μ experimentally I go to the lab and insert the rod within a solenoid of many turns having a diameter exactly matching the diameter of the rod. I observe that the time constant of exponential decay of the solenoid current in an RL circuit that I construct increases by 0.05% when the rod is inserted to replace the air core of the solenoid. Determine μ in terms of μ_o. Is the rod diamagnetic or paramagnetic? Explain.



$L \propto \mu$ & $\tau \propto L \propto \mu \Rightarrow$ if $\tau \uparrow$ then $\mu \uparrow$

$\hookrightarrow \mu = \left(1 + \frac{0.05}{100}\right)\mu_0 = 1.0005\,\mu_0$ // Paramagnetic

   b) Recall that

$$\gamma\eta = j\omega\mu \quad \text{and} \quad \frac{\gamma}{\eta} = \sigma + j\omega\epsilon$$

for wave propagation in a conducting (lossy) medium where a plane TEM wave with a magnetic field intensity vector

$$\mathbf{H}(z,t) = \hat{x}\,25e^{-z}\cos(8\pi\cdot 10^6 t - \sqrt{3}z - \frac{\pi}{3})\,\frac{\text{A}}{\text{m}}$$

is observed — if $\mu = \mu_o$ in the medium, determine:

   i. (6 pts) Radian wave frequency $\omega = 2\pi f$, the attenuation constant α, wavenumber β, and propagation constant $\gamma = \alpha + j\beta$,

$\omega = 8\pi\times 10^6 \ \frac{\text{rad}}{\text{s}}$

$\alpha = 1 \ \frac{\text{Np}}{\text{m}}$, $\beta = \sqrt{3} \ \frac{\text{rad}}{\text{m}}$, $\gamma = \alpha + j\beta = 1 + j\sqrt{3} \ \frac{1}{\text{m}} = 2\left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = 2e^{j\frac{\pi}{3}} \ \frac{1}{\text{m}}$ ($e^{j\frac{\pi}{3}}$)

   ii. (3 pts) The intrinsic impedance η,

$\eta = \frac{j\omega\mu_0}{\gamma} = j\frac{(8\pi\times 10^6)(4\pi\times 10^{-7})}{2e^{j\pi/3}} = j\frac{16\pi^2}{10}e^{-j\pi/3} = \frac{16\pi^2}{10}e^{j\left(\frac{\pi}{2}-\frac{\pi}{3}\right)} = \frac{8}{5}\pi^2 e^{j\frac{\pi}{6}}\ \Omega$

   iii. (3 pts) The permittivity ϵ,

$\frac{\gamma}{\eta} = \frac{2e^{j\frac{\pi}{3}}}{j\frac{16\pi^2}{10}e^{-j\pi/3}} = -j\frac{5}{4\pi^2}e^{j\left(\frac{2\pi}{3}\right)} \Rightarrow \omega\epsilon = \text{Im}\left[-j\frac{5}{4\pi^2}e^{j\left(\frac{2\pi}{3}\right)}\right]$

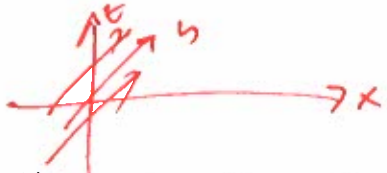
$\epsilon = \frac{-\cos\left(\frac{2\pi}{3}\right)\cdot 5}{(4\pi^2)(8\pi\times 10^{+6})} = \frac{5\times 10^{-6}}{64\pi^3} \cong 2.5\times 10^{-9} \ \frac{\text{F}}{\text{m}}$ ($\cos\left(\frac{2\pi}{3}\right) = -1/2$)

   iv. (1 pt) Phasor $\tilde{\mathbf{H}}$,

$\tilde{H} = \hat{x}\,25\,e^{-z}e^{-j\pi/3}e^{-j\sqrt{3}z} \ \frac{\text{A}}{\text{m}}$

   v. (2 pts) The corresponding phasor $\tilde{\mathbf{E}}$.

$\tilde{E} = -\hat{y}(25)\eta\,e^{-z}e^{j\pi/3}e^{-j\sqrt{3}z} \ \frac{\text{V}}{\text{m}}$



2. Consider a time-varying surface current density $\mathbf{J}_s(t) = u(t)\sin(2\pi ft)\hat{y}$ A/m, with $f = \frac{1}{2} \times 10^6$ Hz, residing on the $x = 0$ surface embedded in free space. In this expression $u(t)$ is the unit-step function modulating the sine signal with a frequency $\omega = 2\pi f$ rad/s.

a) (4 pts) What is the vector wave field $\mathbf{E}(x,t)$ caused by $\mathbf{J}_s(t)$ in the region $x > 0$?

$$\bar{E}(x,t) = -\hat{y}\frac{\eta}{2}u\left(t - \frac{x}{v}\right)\sin\left(2\pi f\left(t - \frac{x}{v}\right)\right)$$

b) (4 pts) What is the vector wavefield $\mathbf{H}(x,t)$ in the region $x < 0$?

$$\bar{H}(x,t) = +\hat{z}\frac{1}{2}u\left(t + \frac{x}{v}\right)\sin\left(2\pi f\left(t + \frac{x}{v}\right)\right)$$

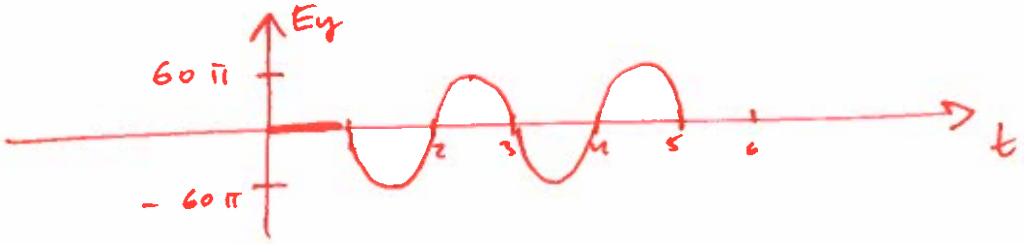
c) (3 pts) What is the period $T = \frac{2\pi}{\omega}$ of the modulated $\mathbf{E}(x,t)$ and $\mathbf{H}(x,t)$ waveforms in μs units ?

$$T = \frac{2\pi}{\omega} = \frac{1}{f} = \frac{1}{\frac{1}{2}\times 10^6} = 2\times 10^{-6}\,s = 2\ \mu s$$

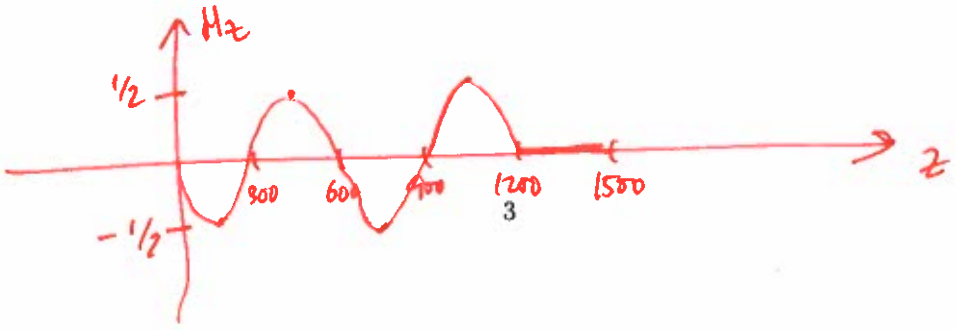
d) (3 pts) What is the wavelength λ of the modulated $\mathbf{E}(x,t)$ and $\mathbf{H}(x,t)$ waveforms?

$$\lambda = \frac{v}{f} = \frac{3\times 10^8\ m/s}{\frac{1}{2}\times 10^6\ 1/s} = 6\times 10^2\ m = 600\ m.$$

e) (5 pts) Plot $E_y(x,t)$ vs t at $x = 300$ m over the time interval $0 < t < 5$ μs. Label both axes carefully. **Hint:** E_y is initially zero until the wavefront arrives at 300 m.

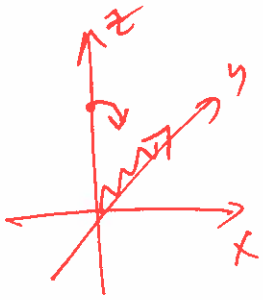


f) (6 pts) Plot $H_z(x,t)$ vs x at $t = 4\,\mu s$ over the region $0 < x < 1500$ m. Label both axes carefully. **Hint:** H_z is zero at distances where the wavefront has not yet reached.



3. This problem has two independent parts:

a) Consider a *right*-circular-polarized TEM plane wave in free-space propagating in y direction. The instantaneous electric field vector of the wave is $2\hat{z}$ V/m at time $t = 0$ at the location $(x, y, z) = (0, 0, 0)$. The wave frequency is $f = 150$ MHz. If the wave electric field phasor is denoted as



$$\tilde{\mathbf{E}} = (A\hat{x} + B\hat{y} + C\hat{z})e^{j(ax+by+cz)} \frac{\text{V}}{\text{m}}$$

determine:

i. (4 pts) The numerical values of constants a, b, and c. **Hint:** what is β?

$f = 150\times10^6$ Hz $\Rightarrow \lambda = \frac{c}{f} = \frac{300\times10^6}{150\times10^6} = 2$ m $\Rightarrow \beta = \frac{2\pi}{\lambda} = \pi$

$a = 0 = c$, $b = -\pi$ rad/m

ii. (4 pts) The numerical values of constants A, B, and C.

$B = 0$, $A = -j2\frac{V}{m}$, $C = 2\frac{V}{m}$ (so that $\tilde{E}(0,0) = 2\hat{z}$ V/m)

iii. (4 pts) The time-averaged Poynting vector $\langle \mathbf{E}\times\mathbf{H} \rangle$ — simplify without using your calculator. **Hint:** twice the linear polarized value

$\langle \tilde{E}\times\tilde{H}^* \rangle = 2\times\frac{E_z^2}{2\eta_0} = \frac{4}{120\pi} = \frac{1}{30\pi}\frac{W}{m^2}$

Circular is twice linear in power!

b) Consider TEM wave propagation in $-z$ direction in some homogeneous medium with $\mu = \mu_o$, $\epsilon = 81\epsilon_o$, $\sigma = 4$ S/m — the wave frequency is $f = 1$ kHz and the wavelength is longer than the penetration depth.

i. (4 pts) Does the medium behave as an *imperfect dielectric* or else as a *good conductor* at the given wave frequency? Explain why.

→ because in imperfect dielectric wavelength is much shorter than the penetration distance but this one is reverse!!

ii. (2 pts) What is the numerical value of the penetration depth δ — simplify without using calculator?

$\delta = \frac{1}{\sqrt{\pi f \mu_o \sigma}} = \frac{1}{\sqrt{\pi\, 10^3 4\pi\times10^{-7} 4}} = \frac{10^2}{4\pi} = \frac{25}{\pi}$ m

iii. (2 pts) What is the numerical value of wavelength λ — simplify without using calculator?

$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\alpha} = 2\pi\frac{25}{\pi} = 50$ m

iv. (2 pts) What is the numerical value of the propagation velocity $v_p = \omega/\beta$ — simplify without using calculator?

$v_p = \frac{\omega}{\beta} = \frac{2\pi f}{2\pi/\lambda} = \lambda f = (50\text{ m})(10^3\frac{1}{s}) = 50\times10^3\frac{m}{s} = 50\frac{km}{s}$

v. (3 pts) If the wave is transporting 1 W/m^2 power density at $z = 0$ what is the power density transported at $z = -10$ m? — you can use your calculator in this part.

$S(-10) = S(0)e^{-2\alpha 10}$, $\frac{1}{\delta} = e^{-\frac{20\pi}{25}} = e^{-4\pi/5} \cong 0.081$ W/m^2

(S(0) = 1 W/m^2)